AI Email Inquiry Manager

Autonomous Communications & Lead Capture System Architecture

PRODUCTION BLUEPRINT

Platform: Python 3.10+ / Google APIs | **AI Engine:** Google Gemini 2.5 Flash | **Integrations:** Gmail & Google Sheets | **Target Brand:** Sam Jewels

1. END-TO-END PIPELINE ARCHITECTURE

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Inbox Trigger	**Daemon Filter**	**AI Core (Gemini)**	**Sheets CRM**	**Instant Reply**
Polls unread buyer inquiries from Gmail via OAuth2	Discards no-reply, billing, and system alert emails	Extracts buyer entities & intent via strict JSON schema	Appends structured record to centralized ledger	Dispatches styled HTML reply directly in the thread

2. INGESTION & INBOX MONITORING

- **Automated Gmail Inbox Polling:** Connects securely using official Google API (`gmail/v1`) with OAuth 2.0, querying for unread buyer inquiries (`is:unread label:INBOX`).
- **Intelligent Daemon & Spam Filtering:** Utilizes an active exclusion list (`IGNORE_KEYWORDS`) to automatically skip and silence non-actionable emails (`no-reply` , `alerts@` , `mailer-daemon` , `donotreply` , `google.com` , etc.).
- **Zero-Duplicate Archiving:** Once an inquiry is handled, the pipeline automatically strips the `UNREAD` label to prevent reprocessing loops and keep your inbox organized.

3. AI INTELLIGENCE & LEAD QUALIFICATION

- **Powered by Gemini 2.5 Flash:** Ultra-fast reasoning and contextual understanding on unstructured email messages, subject lines, and sender signatures.
- **Strict Pydantic JSON Schema:** Guarantees zero hallucinations through typed schema validation:
  - `sender_name` : Full prospect name extracted from greeting, body, or metadata.
  - `company_name` : Detected business or firm (defaults to 'Individual').
  - `inquiry_category` : Categorized into *Enterprise Sales*, *Product Support*, *Partnership*, or *General Inquiry*.
  - `urgency_level` : Evaluated priority grade (*High*, *Medium*, or *Low*).
  - `summary_intent` : Concise 1-sentence breakdown of what the prospect needs.
  - `recommended_reply` : Context-aware personalized reply draft addressing specific inquiries.
- **High-Demand Resilience (503 Auto-Retry):** Handles transient API throttling with automated exponential backoff up to 5 successive attempts.

4. CENTRALIZED GOOGLE SHEETS CRM SYNC

Every qualified lead is logged in real-time into the target destination ledger (`Lead_Inquiries_2026`):

Column	Field Name	Description
Column A	`Timestamp`	Execution timestamp in ISO 8601 UTC format.
Column B	`Sender Name`	Extracted full name of the prospect.
Column C	`Sender Email`	Direct contact email address parsed from header.
Column D	`Company`	Identified enterprise, organization, or personal client.
Column E	`Category`	Intent category (Sales, Support, Partnership, General).
Column F	`Summary / Intent`	Synthesized 1-sentence requirement breakdown.

5. INSTANT CONTEXT-AWARE AUTO-REPLY DISPATCHER

- **Brand Voice & Persona:** Configured for **Sam Jewels** with strict professional standards: double line-breaks after greeting, 1–2 focused readable paragraphs, and clean corporate sign-offs.
- **Clean HTML Styling:** Automatically converts raw text newlines into web-safe HTML formatting (`Arial` , 14px font, 1.6 line height) to render consistently on mobile and desktop email clients.
- **Thread Continuity (`threadId`):** Replies directly inside the customer's existing thread, preserving full context and subject history (`Re: [Subject]`).

6. EXECUTION MODES & DEVELOPER TOOLING

Execution Mode	Command	Functionality
Full Production	`python pipeline.py`	Autonomous live pipeline: polls Gmail, parses with Gemini, appends to Sheets, sends replies.
Dry Run Mode	`python pipeline.py --dry-run`	Inspects live Gmail messages & tests Gemini extraction without writing rows or sending replies.
Local Simulation	`python pipeline.py --simulate`	Tests full extraction, schema mapping, and output formatting offline with zero credentials.
1-Click OAuth Setup	`python setup_auth.py`	Spawns local OAuth server & browser to generate authorized `token.json` credentials.

7. INTERACTIVE WEB PROTOTYPE & ENTERPRISE SECURITY

- **Interactive UI Simulation (`index.html`):** Self-contained Tailwind CSS dashboard with preset scenarios, stage visualizers, live client-side Gemini API inference, and 1-click CSV export.
- **Zero-Leakage Git Protection:** Enforced through `.gitignore` , strictly barring sensitive tokens (`credentials.json` , `token.json` , and `.env`) from ever being committed to GitHub.

- **Automatic OAuth Refresh:** Seamlessly renews expired Google access tokens in the background without requiring user re-authentication.
